

1.6

69

Let S_1 and S_2 be finite subsets of R^n such that S_1 is contained in S_2 . Prove that if S_1 is a generating set for R^n , then so is S_2 .

74

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70

Let $\mathbf{u}$ and $\mathbf{v}$ be any vectors in R^n . Prove that the spans of $\{\mathbf{u}, \mathbf{v}\}$ and $\{\mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v}\}$ are equal.

71

Let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ be vectors in R^n and $c_1, c_2, \dots, c_k$ be nonzero scalars. Prove that $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\} = \text{Span}\{c_1\mathbf{u}_1, c_2\mathbf{u}_2, \dots, c_k\mathbf{u}_k\}$

72

Let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ be vectors in R^n and c be a scalar. Prove that the span of $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is equal to the span of $\{\mathbf{u}_1 + c\mathbf{u}_2, \mathbf{u}_2, \dots, \mathbf{u}_k\}$.

77

Let B be a matrix obtained from A by performing a single elementary row operation on A . Prove that the span of the rows of B equals the span of the rows of A .

- 1. Interchange: Interchange any two rows of the matrix
- 2. Scaling: Multiply every entry of some row by the same nonzero scalar
- 3. Row Addition: Add a multiple of one row of the matrix to another row

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78

Prove that every linear combination of the rows of A can be written as a linear combination of the rows of the reduced row echelon form of A .

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